Decentralized Algorithm for Joint Data Placement and Rate Allocation in Content-Aware Networks

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Abstract. In this paper a mathematical model of data delivery in Content-Aware Network (CAN) is studied. Such network architecture assumes that users access data objects (content) replicated on a number of cache servers. A CAN operator decides on the placement of data objects on behalf of the content publishers, and decides on the transmission rate allocation of users (content consumers downloading content, and publishers uploading and updating their content). A problem of minimizing the mean transmission delay is considered and formulated as a mixed-integer programming problem. Decomposition-based heuristic is proposed, which utilizes projected gradient descent and greedy strategy. The algorithm is presented as a decentralized protocol, embedded in the network architecture.

Keywords: Content-Aware Network, flow control, facility location.

1 Introduction

The problem of optimal planning of data delivery networks in the Internet has been studied systematically in the last decade, in response to rapidly increasing popularity of the Web [1]. The various formulations of placement problems of servers, proxies and caches in computer networks have been studied in e.g.: [2,3]. The first Content Delivery Network was deployed by Akamai Technologies, as a result of the pioneering research in this area [4]. Since then, most of the major content providers entrusted their offered services to such systems. Recently, several new system architectures for future Internet have been proposed, such as Content Centric Network [5] and Content-Aware Networks, optimized for delivery of large volumes of content [6].

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In this paper a mathematical model of Content-Aware Network (CAN) is presented. In such network the transit routers are supplied with special cache servers, which store the content replicas generated on behalf of the content publishers. The performance of the considered system is expressed as a mean transmission delay between users and servers. A decomposition-based heuristic algorithm is presented, which allows for sequential solving the rate allocation [7,8] and content placement [9,10] problems, utilizing projected gradient descent and greedy strategy, respectively.

2 Problem Statement

In the considered problem given is a set of users (called *clients*) who want to read data objects and a set of users (called *publishers*) who want to have their data content delivered to the clients. Clients never read data directly from the publishers' servers. Instead, publishers place the data objects inside public cache servers, connected to the network transit routers. These cache servers are under control of network *operator*, who manages the placement of content on behalf of publishers, as well as assigns transmission rates. Further in this paper the term *user* refers to the local area network (LAN) which generates a packet *flow*, and can be either client user LAN or publisher user LAN.

The description above illustrates the concept of data delivery in *Content-Aware Networks*. The following decisions are to be made by the network operator:

1. how to allocate transmission rates to the users (clients and publishers),
2. in which cache servers to place individual data objects,
3. how to redirect clients to the cache servers.

All those decisions influence the system's performance in terms of request delays perceived by users (clients and publishers). It is assumed that the objective of network operator's decisions is to minimize the sum of mean delays. The formulation is presented in the next subsections and the summary of notation is contained in Table 1.

2.1 Objective

The total delay, associated with mth user's requests concerning nth object on server s, is denoted by $\tau_{mns}(\mathbf{x}, \mathbf{y})$ and it can be given with the formula:

$$\tau_{mns}(\mathbf{x}, \mathbf{y}) = \tau^{(1)}_{mns}(y_{mns}) + \sum_{l=1}^{L} a_{mnsl} \left(\tau^{(2)}_{l}(\mathbf{x}, \mathbf{y}) + \tau^{(3)}_{l} \right)$$

where $\tau^{(1)}_{mns}(y_{mns})$ is a time it takes to push all nth object's bits onto link by the source node, $\tau^{(2)}_{l}(\mathbf{x}, \mathbf{y})$ is a queue sojourn delay (incorporating packet's waiting and serving time on the router's interface), $\tau^{(3)}_{l}$ are the rest of constant delays

Table 1. Summary of notation

Symbol	Description
$m \in \{0, 1, \ldots, M\}$	user's index, where $m = 0$ denotes publisher and $m \in \{1, \ldots M\}$ denotes client,
$n \in \{1, \ldots, N\}$	data object's index,
$s \in \{1, \ldots, S\}$	cache server's index,
$l \in \{1, \ldots, L\}$	transmission link's index,
$x_{mns} \in \{0, 1\}$	assignment of m-th client to s-th server for reading object n,
$x_{0ns} \in \{0, 1\}$	assignment of n-th object the cache server s $x_{ns} = [x_{0ns}]$, $n = 1, 2, \ldots, N, s = 1, 2, \ldots, S$, $\mathbf{x} = [x_{mns}]$, $m = 0, 1, \ldots, M, n = 1, 2, \ldots, N, s = 1, 2, \ldots, S$,
$y_{mns} \geq 0$	rate allocated to the m-th client for downloading n-th object from server s,
$y_{0ns} \geq 0$	rate allocated to the publisher of n-th object for uploading this object to server s, $\mathbf{y} = [y_{mns}]$, $m = 0, 1, \ldots, M, n = 1, 2, \ldots, N, s = 1, 2, \ldots, S$,
$y_{mns,\max} \geq 0$	maximum achievable rate for transmitting n-th object from/to m-th user to/from s-th server; here it is assumed to equal: $y_{mns,\max} = \min_{l \in \{l: a_{mnsl}=1\}} C_l$,
$a_{mnsl} \in \{0, 1\}$	routing variable, which defines if user m uses link l while downloading (uploading) object n from (to) server s,
C_l	capacity of physical link l,
b_n	size of object n

related to the link l (e.g. processing delay, propagation delay). So, the objective function is given as:

$$Q(\mathbf{x}, \mathbf{y}) = \sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} x_{mns} \tau_{mns}(\mathbf{x}, \mathbf{y}).$$

2.2 Constraints

The following requirements imposed by the problem may be formulated:

1. Each user must be able to download every object from one location:

$$\forall_{m \in \{1,\ldots,M\}} \forall_{n=1,\ldots,N} \quad \sum_{s=1}^{S} x_{mns} = 1 \ . \tag{1}$$

2. There must be a copy of an object located on the server for which this user is assigned to:

$$\forall_{m \in \{1,\ldots,M\}} \forall_n \forall_s \quad x_{mns} \leq x_{0ns} \ . \tag{2}$$

3. The links' capacities must not be exceeded:

$$\forall_{l=1,\ldots,L} \quad \sum_{m=0}^{M} \sum_{n=1}^{N} \sum_{s=1}^{S} a_{mnsl} x_{mns} y_{mns} \leq C_l \ . \tag{3}$$

Observe that constraints (1) and (2) imply that at least one copy of each object must be located in the network, i.e. $\forall_{n=1,\ldots,N} \quad \sum_{s=1}^{S} x_{0ns} \geq 1$. In this paper, it is assumed, that storage capacity on every server is big enough to keep all the objects (i.e. $\forall_{s\in\{1,\ldots,S\}} \sum_{n=1}^{N} b_n \leq B_s$, where B_s is the storage capacity of server s). Without such assumption the problem becomes much more difficult and hard to approximate [11], however it appears that mass storage is the least deficient resource in practice.

2.3 Problem Formulation

The joint object placement and capacity allocation problem in Content-Aware Network is formulated as follows. For given values of $M, N, S, L, \mathbf{A} = [a_{mnsl}]$, $\mathbf{C} = [C_l]$ one should find:

$$(\mathbf{x}^*, \mathbf{y}^*) = \arg\min_{\mathbf{x},\mathbf{y}} Q(\mathbf{x}, \mathbf{y})$$

such that constraints (1)–(3) hold.

This is nonlinear mixed integer programming problem (discrete-continuous). Even assuming that $\mathbf{y}^*$ is known in advance the problem is NP-hard with respect of $\mathbf{x}$. To obtain exact solution, algorithms such as branch-and-bound or outer approximation [12] may be applied. However, such approach leads to solving procedures with prohibitive running time (requiring exponential number of steps in terms of M, N, S and L). Moreover, these procedures are hard to implement effectively in decentralized manner. In this paper, a particular heuristic approach based on problem decomposition is proposed and its decentralized version is presented. The general formulation of objective functions allows for plugging functions representing the time spent by packets inside routers' queues arbitrarily. However, in this paper it is assumed that all routers adhere to M/M/1 queueing characteristic, i.e. mean sojourn time of packet in link's l router is $\tau_l^{(2)}(\mathbf{x}, \mathbf{y}) \approx (C_l - \sum_{m,n,s} x_{mns} y_{mns})^{-1}$. This is justified for FTP-type transactions, and it is assumed that such traffic dominates the transmissions in Content-Aware Networks. Thus solving the problem formulated in this paper can be seen as optimizing the network with respect to the static file transfer applications. For other types of traffic different queueing models need to be taken into consideration [13].

3 Solution Algorithms

An additional specificity of the problem is the decentralized architecture of the system. It is required that the solution algorithms can be distributed across the network devices, since both input data, as well as decision variables, are bound to spatially distributed network nodes. The implementation should assume the limited communication between entities cooperating in the computation of the global solution. In this section the idea of decomposition is presented, followed by the description of two-level solution algorithm, and its implementation in the decentralized model. Such implementation consists of routines which must be implemented on specified network nodes, communicating via fixed protocol.